

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12401627
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Crabtree; Jason et al.

System and methods for predictive cyber-physical resource management

Abstract

A system and method for predictive cyber-physical resource management, including a business operating system, parameter evaluation engine, at least one cyber-physical asset, at least one crypt-ledger, a network, and the ability to represent data in Markov State Models and finite state machines.

Inventors:	Crabtree; Jason (Vienna, VA), Kelley; Richard (Woodbridge, VA)
Applicant:	QOMPLX LLC (Reston, VA)
Family ID:	1000008778348
Assignee:	QOMPLX LLC (Reston, VA)
Appl. No.:	18/304343
Filed:	April 21, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230388277 A1	Nov. 30, 2023

Related U.S. Application Data

continuation parent-doc US 15904006 20180223 US 10652219 20200512 child-doc US 15930063
continuation-in-part parent-doc US 18172213 20230221 ABANDONED child-doc US 18304343
continuation-in-part parent-doc US 15930063 20200512 US 11588793 20230221 child-doc US
18172213
continuation-in-part parent-doc US 15725274 20171004 US 10609079 20200331 child-doc US
15904006
continuation-in-part parent-doc US 15655113 20170720 US 10735456 20200804 child-doc US

15725274
continuation-in-part parent-doc US 15616427 20170607 ABANDONED child-doc US 15655113
continuation-in-part parent-doc US 15237625 20160815 US 10248910 20190402 child-doc US 15655113
continuation-in-part parent-doc US 15206195 20160708 ABANDONED child-doc US 15616427
continuation-in-part parent-doc US 15186453 20160618 ABANDONED child-doc US 15206195
continuation-in-part parent-doc US 15166158 20160526 ABANDONED child-doc US 15186453
continuation-in-part parent-doc US 15141752 20160428 US 10860962 20201208 child-doc US 15166158
continuation-in-part parent-doc US 15091563 20160405 US 10204147 20190212 child-doc US 15141752
continuation-in-part parent-doc US 14986536 20151231 US 10210255 20190219 child-doc US 15141752
continuation-in-part parent-doc US 14925974 20151028 ABANDONED child-doc US 15141752
continuation-in-part parent-doc US 14925974 20151028 ABANDONED child-doc US 15616427

Publication Classification

Int. Cl.: **H04L9/40** (20220101); **G06F16/951** (20190101); **G06N7/01** (20230101); **H04L9/14** (20060101); **H04L9/32** (20060101); G06N5/01 (20230101); G06N5/045 (20230101); G06N5/046 (20230101); G06N20/00 (20190101); H04L9/00 (20220101)

U.S. Cl.:

CPC **H04L63/0428** (20130101); **G06F16/951** (20190101); **G06N7/01** (20230101); **H04L9/14** (20130101); **H04L9/3236** (20130101); **H04L9/3297** (20130101); **H04L63/061** (20130101); **H04L63/1408** (20130101); G06N5/01 (20230101); G06N5/045 (20130101); G06N5/046 (20130101); G06N20/00 (20190101); H04L9/50 (20220501); H04L63/0442 (20130101); H04L63/123 (20130101)

Field of Classification Search

CPC: H04L (63/0428); H04L (9/14); H04L (9/3236); H04L (9/3297); H04L (63/061); H04L (63/1408); H04L (63/0442); H04L (63/123); H04L (2209/38); G06F (16/951); G06N (7/005); G06N (5/003); G06N (5/045); G06N (5/046); G06N (20/00)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6256544	12/2000	Weissinger	N/A	N/A
8788254	12/2013	Peloski	N/A	N/A
9020802	12/2014	Florissi et al.	N/A	N/A
9122694	12/2014	Dukes et al.	N/A	N/A
9299029	12/2015	Kim	N/A	N/A
9350550	12/2015	Nix	N/A	N/A
9357381	12/2015	Cho et al.	N/A	N/A
9426118	12/2015	Kim	N/A	N/A
9451462	12/2015	Kim et al.	N/A	N/A

9602530	12/2016	Ellis et al.	N/A	N/A
9967334	12/2017	Ford	N/A	N/A
10061635	12/2017	Ellwein	N/A	N/A
10200233	12/2018	Anderson et al.	N/A	N/A
10248910	12/2018	Crabtree et al.	N/A	N/A
10262321	12/2018	Ramanathan et al.	N/A	N/A
10320813	12/2018	Ahmed	N/A	H04L 63/1416
10367829	12/2018	Huang et al.	N/A	N/A
10469329	12/2018	Eicher et al.	N/A	N/A
10540202	12/2019	Smaldone et al.	N/A	N/A
11017096	12/2020	Tsokos et al.	N/A	N/A
2007/0018953	12/2006	Kipersztok	N/A	N/A
2007/0203693	12/2006	Estes	N/A	N/A
2012/0197911	12/2011	Banka	707/754	G06F 16/951
2013/0104236	12/2012	Ray	726/25	H04L 63/20
2014/0351642	12/2013	Bates et al.	N/A	N/A
2014/0358911	12/2013	McCarthy et al.	N/A	N/A
2015/0371224	12/2014	Lingappa	N/A	N/A
2015/0379072	12/2014	Dirac et al.	N/A	N/A
2015/0379510	12/2014	Smith	705/71	G06F 21/64
2016/0275123	12/2015	Lin et al.	N/A	N/A
2016/0379224	12/2015	Cecchi et al.	N/A	N/A
2019/0132344	12/2018	Lem et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2014159150	12/2013	WO	N/A
2017075543	12/2016	WO	N/A

Primary Examiner: Huang; Cheng-Feng

Attorney, Agent or Firm: Galvin Patent Law LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) Priority is claimed in the application data sheet to the following patents or patent applications, each of which is expressly incorporated herein by reference in its entirety: Ser. No. 18/172,213 Ser. No. 15/930,063 Ser. No. 15/904,006 Ser. No. 15/725,274 Ser. No. 15/655,113 Ser. No. 15/616,427 Ser. No. 14/925,974 Ser. No. 15/237,625 Ser. No. 15/206,195 Ser. No. 15/186,453 Ser. No. 15/166,158 Ser. No. 15/141,752 Ser. No. 15/091,563 Ser. No. 14/986,536

BACKGROUND OF THE INVENTION

Field of the Art

(2) The disclosure relates to the field of network enabled asset tracking and management, more specifically to the field of managing inventory assets as part of technology operations and risk to include cybersecurity.

Discussion of the State of the Art

(3) Currently, it is possible for corporations and individuals to track certain assets in certain ways, to ensure their safety and ensure valid operation. For example, it is possible to track packages

shipped via many shipping corporations, and it is possible and commonplace to have temperature controls and monitoring in certain environments such as museums, libraries and wine cellars. However, the breadth and depth of sensor monitoring with cyber-enabled physical assets is low, and the cost and barrier of entry into such endeavors is usually quite high. What's more, in some situations, again referring to tracking shipping packages or less than truckload freight, one is not necessarily relying not the same sensor to report the location of the package in question and its environment. Rather, one is often relying on human operators or separate business systems to reliably report the presence of the object at or between disparate facilities. Human interaction, and therefore human error, is present alongside a pressing data integration and validity challenge in many of these situations and more, such as the issue of contractual obligations involving the physical status of objects or areas, which must be enforced by other humans, and therefore involve unknown variables into the business transaction.

(4) What is needed is a system and methods for autonomous sensor data monitoring of multiple various forms of cyber-physical assets, to improve operational risk management to include asset management and supply chain risk management, and engaging these assets in the use of a decision platform which may also optionally include decentralized decision-making or actions using self-fulfilling smart contracts.

SUMMARY OF THE INVENTION

(5) Accordingly, the inventor has conceived and reduced to practice, in a preferred embodiment of the invention, a system and methods for predictive cyber-physical resource management. The following non-limiting summary of the invention is provided for clarity and should be construed consistently with embodiments described in the detailed description below.

(6) According to a preferred aspect, a system for predictive cyber-physical resource management, comprising: a first computing device comprising a first processor, a first memory, a wired or wireless communication device, and a first plurality of programming instructions stored in the first memory and operating on the first processor, wherein the first plurality of programmable instructions, when operating on the first processor, cause the computing device to: receive status update messages from one or more physical assets; for each physical asset for which a status message is received, update a current state record for the physical asset based on the content of the status message; append the current state record to a time-series record of the asset's state over time; append the interactions of the asset with other assets and people and places in a graph database; where a status update message for a particular physical asset is not received, apply a machine learning algorithm to the state model to determine a predicted state of the particular physical asset; append the predicted state to a time-series record of the particular asset's state over time; and, monitor the state and structure of the interaction graph between the asset (and optionally similar assets or classes of assets) via comparison of timeseries representations of the asset graph and interaction graph, is disclosed.

(7) According to another preferred aspect, a method for predictive cyber-physical resource management, comprising the steps of: receiving, at a first computing device, status update messages from one or more physical assets; for each physical asset for which a status message is received, updating a current state record for the physical asset based on the content of the status message; appending the current state record to a time-series record of the asset's state over time; where a status update message for a particular physical asset is not received, applying a machine learning algorithm to the state model to determine a predicted state of the particular physical asset; appending the predicted state to a time-series record of the particular asset's state over time; and, appending the predicted asset graph or interaction graph to a temporally enriched graph record of the particular asset's interaction and state over time, is disclosed.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

- (1) The accompanying drawings illustrate several aspects and, together with the description, serve to explain the principles of the invention according to the aspects. It will be appreciated by one skilled in the art that the particular arrangements illustrated in the drawings are merely exemplary and are not to be considered as limiting of the scope of the invention or the claims herein in any way.
- (2) FIG. 1 is a diagram of an exemplary architecture of a system for the capture and storage of time series data from sensors with heterogeneous reporting profiles according to a preferred aspect of the invention.
- (3) FIG. 2 is a diagram of an exemplary architecture of a business operating system according to a preferred aspect of the invention.
- (4) FIG. 3 is a diagram of an exemplary architecture of an automated planning service cluster and related modules according to a preferred aspect.
- (5) FIG. 4 is a system diagram illustrating connections between core components of the invention for geo-locating and tracking the status of cyber-physical assets, according to a preferred aspect.
- (6) FIG. 5 is a method diagram illustrating key steps in the communication between cyber-physical assets and remote servers, according to a preferred aspect.
- (7) FIG. 6 is a method diagram illustrating key steps in a business operating system interacting with data received from cyber-physical assets in databases to verify updates in a cryptographic ledger, according to a preferred aspect.
- (8) FIG. 7 is a method diagram illustrating several steps in the use of smart contracts combined with cyber-physical assets, according to a preferred aspect.
- (9) FIG. 8 is a method diagram illustrating key steps in the function of a parametric evaluation engine, according to a preferred aspect.
- (10) FIG. 9 is a block diagram illustrating an exemplary hardware architecture of a computing device.
- (11) FIG. 10 is a block diagram illustrating an exemplary logical architecture for a client device.
- (12) FIG. 11 is a block diagram showing an exemplary architectural arrangement of clients, servers, and external services.
- (13) FIG. 12 is another block diagram illustrating an exemplary hardware architecture of a computing device.
- (14) FIG. 13 is a method diagram illustrating an exemplary process for using status update messages to predict the status of network resources and continuously refine predictive logic, according to a preferred aspect.
- (15) FIG. 14 is a method diagram illustrating an exemplary method for scheduling network tasks based on predicted availability and conditions, according to a preferred aspect.

DETAILED DESCRIPTION OF THE DRAWING FIGURES

- (16) The inventor has conceived, and reduced to practice, a system and methods for predictive cyber-physical resource and risk management.
- (17) One or more different aspects may be described in the present application. Further, for one or more of the aspects described herein, numerous alternative arrangements may be described; it should be appreciated that these are presented for illustrative purposes only and are not limiting of the aspects contained herein or the claims presented herein in any way. One or more of the arrangements may be widely applicable to numerous aspects, as may be readily apparent from the disclosure. In general, arrangements are described in sufficient detail to enable those skilled in the art to practice one or more of the aspects, and it should be appreciated that other arrangements may be utilized and that structural, logical, software, electrical and other changes may be made without departing from the scope of the particular aspects. Particular features of one or more of the aspects described herein may be described with reference to one or more particular aspects or figures that

form a part of the present disclosure, and in which are shown, by way of illustration, specific arrangements of one or more of the aspects. It should be appreciated, however, that such features are not limited to usage in the one or more particular aspects or figures with reference to which they are described. The present disclosure is neither a literal description of all arrangements of one or more of the aspects nor a listing of features of one or more of the aspects that must be present in all arrangements.

(18) Headings of sections provided in this patent application and the title of this patent application are for convenience only and are not to be taken as limiting the disclosure in any way.

(19) Devices that are in communication with each other need not be in continuous communication with each other, unless expressly specified otherwise. In addition, devices that are in communication with each other may communicate directly or indirectly through one or more communication means or intermediaries, logical or physical.

(20) A description of an aspect with several components in communication with each other does not imply that all such components are required. To the contrary, a variety of optional components may be described to illustrate a wide variety of possible aspects and in order to more fully illustrate one or more aspects. Similarly, although process steps, method steps, algorithms or the like may be described in a sequential order, such processes, methods and algorithms may generally be configured to work in alternate orders, unless specifically stated to the contrary. In other words, any sequence or order of steps that may be described in this patent application does not, in and of itself, indicate a requirement that the steps be performed in that order. The steps of described processes may be performed in any order practical. Further, some steps may be performed simultaneously despite being described or implied as occurring non-simultaneously (e.g., because one step is described after the other step). Moreover, the illustration of a process by its depiction in a drawing does not imply that the illustrated process is exclusive of other variations and modifications thereto, does not imply that the illustrated process or any of its steps are necessary to one or more of the aspects, and does not imply that the illustrated process is preferred. Also, steps are generally described once per aspect, but this does not mean they must occur once, or that they may only occur once each time a process, method, or algorithm is carried out or executed. Some steps may be omitted in some aspects or some occurrences, or some steps may be executed more than once in a given aspect or occurrence.

(21) When a single device or article is described herein, it will be readily apparent that more than one device or article may be used in place of a single device or article. Similarly, where more than one device or article is described herein, it will be readily apparent that a single device or article may be used in place of the more than one device or article.

(22) The functionality or the features of a device may be alternatively embodied by one or more other devices that are not explicitly described as having such functionality or features. Thus, other aspects need not include the device itself.

(23) Techniques and mechanisms described or referenced herein will sometimes be described in singular form for clarity. However, it should be appreciated that particular aspects may include multiple iterations of a technique or multiple instantiations of a mechanism unless noted otherwise. Process descriptions or blocks in figures should be understood as representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process. Alternate implementations are included within the scope of various aspects in which, for example, functions may be executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those having ordinary skill in the art.

Definitions

(24) As used herein, a “swimlane” is a communication channel between a time series sensor data reception and apportioning device and a data store meant to hold the apportioned data time series sensor data. A swimlane is able to move a specific, finite amount of data between the two devices.

For example a single swimlane might reliably carry and have incorporated into the data store, the data equivalent of 5 seconds worth of data from 10 sensors in 5 seconds, this being its capacity. Attempts to place 5 seconds worth of data received from 6 sensors using one swimlane would result in data loss. In one common preferred embodiment, a swimlane might be used for a single sensor and have a bijective association with a vertex in the graph database. Additionally a subgraph of the graph, could commonly relate to a metaswimlane as described herein or may trigger a query-based or separate analytically derived set of metrics which may themselves be stored in a swimlane associated with the subgraph (e.g. betweenness, centrality, or other kinds of structural metrics describing the subgraph which may change over time alongside its structure).

(25) As used herein, a “metaswimlane” is an as-needed logical combination of transfer capacity of two or more real swimlanes that is transparent to the requesting process. Sensor studies where the amount of data received per unit time is expected to be highly heterogeneous over time may be initiated to use metaswimlanes. Using the example used above that a single real swimlane can transfer and incorporate the 5 seconds worth of data of 10 sensors without data loss, the sudden receipt of incoming sensor data from 13 sensors during a 5 second interval would cause the system to create a two swimlane metaswimlane to accommodate the standard 10 sensors of data in one real swimlane and the 3 sensor data overage in the second, transparently added real swimlane, however no changes to the data receipt logic would be needed as the data reception and apportionment device would add the additional real swimlane transparently.

(26) Conceptual Architecture

(27) FIG. 1 is a diagram of an exemplary architecture of a system for the capture and storage of time series data from sensors with heterogeneous reporting profiles according to a preferred aspect of the invention. In this embodiment, a plurality of sensor devices **110a-n** stream data to a collection device, in this case a web server acting as a network gateway **115**. These sensors **110a-n** can be of several forms, some non-exhaustive examples being: physical sensors measuring humidity, pressure, temperature, orientation, and presence of a gas; or virtual such as programming measuring a level of network traffic, memory usage in a controller, and number of times the word “refill” is used in a stream of email messages on a particular network segment, to name a small few of the many diverse forms known to the art. In the embodiment, the sensor data is passed without transformation to the data management engine **120**, where it is aggregated and organized for storage in a specific type of data store **125** designed to handle the multidimensional time series data resultant from sensor data. Raw sensor data can exhibit highly different delivery characteristics. Some sensor sets may deliver low to moderate volumes of data continuously. It would be infeasible to attempt to store the data in this continuous fashion to a data store as attempting to assign identifying keys and the to store real time data from multiple sensors would invariably lead to significant data loss. In this circumstance, the data stream management engine **120** would hold incoming data in memory, keeping only the parameters, or “dimensions” from within the larger sensor stream that are pre-decided by the administrator of the study as important and instructions to store them transmitted from the administration device **112**. The data stream management engine **120** would then aggregate the data from multiple individual sensors and apportion that data at a predetermined interval, for example, every 10 seconds, using the timestamp as the key when storing the data to a multidimensional time series data store over a single swimlane of sufficient size. This highly ordered delivery of a foreseeable amount of data per unit time is particularly amenable to data capture and storage but patterns where delivery of data from sensors occurs irregularly and the amount of data is extremely heterogeneous are quite prevalent. In these situations, the data stream management engine cannot successfully use strictly single time interval over a single swimlane mode of data storage. In addition to the single time interval method the invention also can make use of event based storage triggers where a predetermined number of data receipt events, as set at the administration device **112**, triggers transfer of a data block consisting of the apportioned number of events as one dimension and a number of sensor ids as the other. In the

embodiment, the system time at commitment or a time stamp that is part of the sensor data received is used as the key for the data block value of the value-key pair. The invention can also accept a raw data stream with commitment occurring when the accumulated stream data reaches a predesigned size set at the administration device **112**.

(28) It is also likely that that during times of heavy reporting from a moderate to large array of sensors, the instantaneous load of data to be committed will exceed what can be reliably transferred over a single swimlane. The embodiment of the invention can, if capture parameters pre-set at the administration device **112**, combine the data movement capacity of two or more swimlanes, the combined bandwidth dubbed a metaswimlane, transparently to the committing process, to accommodate the influx of data in need of commitment. All sensor data, regardless of delivery circumstances are stored in a multidimensional time series data store **125** which is designed for very low overhead and rapid data storage and minimal maintenance needs to sap resources. The embodiment uses a key-value pair data store examples of which are DynamoDB, Riak, Redis and Berkeley DB for their low overhead and speed, although the invention is not specifically tied to a single data store type to the exclusion of others known in the art should another data store with better response and feature characteristics emerge. Due to factors easily surmised by those knowledgeable in the art, data store commitment reliability is dependent on data store data size under the conditions intrinsic to time series sensor data analysis. The number of data records must be kept relatively low for the herein disclosed purpose. As an example one group of developers restrict the size of their multidimensional time series key-value pair data store to approximately $8.64 \times 10^{sup.4}$ records, equivalent to 24 hours of 1 second interval sensor readings or 60 days of 1 minute interval readings. In this development system the oldest data is deleted from the data store and lost. This loss of data is acceptable under development conditions but in a production environment, the loss of the older data is almost always significant and unacceptable. The invention accounts for this need to retain older data by stipulating that aged data be placed in long term storage. In the embodiment, the archival storage is included **130**. This archival storage might be locally provided by the user, might be cloud based such as that offered by Amazon Web Services (e.g. DocumentDB or S3) or Google or could be any other available very large capacity storage method known to those skilled in the art.

(29) Reliably capturing and storing sensor data as well as providing for longer term, offline, storage of the data, while important, is only an exercise without methods to repetitively retrieve and analyze most likely differing but specific sets of data over time. The invention provides for this requirement with a robust query language that both provides straightforward language to retrieve data sets bounded by multiple parameters, but to then invoke several transformations on that data set prior to output. In the embodiment isolation of desired data sets and transformations applied to that data occurs using pre-defined query commands issued from the administration device **112** and acted upon within the database by the structured query interpreter **135**. Below is a highly simplified example statement to illustrate the method by which a very small number of options that are available using the structured query interpreter **135** might be accessed.

(30) SELECT [STREAMING|EVENTS] data_spec FROM [unit] timestamp TO timestamp
GROUPBY (sensor_id, identifier) FILTER [filter_identifier] FORMAT [sensor [AS identifier] [,
sensor [AS identifier]] . . .] (TEXT|JSON|FUNNEL|KML|GEOJSON|TOPOJSON);

(31) Here “data_spec” might be replaced by a list of individual sensors from a larger array of sensors and each sensor in the list might be given a human readable identifier in the format “sensor AS identifier”. “unit” allows the researcher to assign a periodicity for the sensor data such as second (s), minute (m), hour (h). One or more transformational filters, which include but a not limited to: mean, median, variance, standard deviation, standard linear interpolation, or Kalman filtering and smoothing, may be applied and then data formatted in one or more formats examples of with are text, JSON, KML, GEOJSON and TOPOJSON among others known to the art, depending on the intended use of the data.

(32) FIG. 2 is a diagram of an exemplary architecture of a business operating system **200** according to a preferred aspect. Client access to the system **205** both for system control and for interaction with system output such as automated predictive decision making and planning and alternate pathway simulations, occurs through the system's highly distributed, very high bandwidth cloud interface **210** which is application driven through the use of the Scala web application development environment (e.g. the Typesafe web stack Play with Akka and Kafka) and web interaction operation scaled elastically by services such as AWS ELASTIC BEANSTALK™, both used for standards compliance and ease of development. Much of the business data analyzed by the system both from sources within the confines of the client business, and from cloud-based sources, also enter the system through the cloud interface **210**, data being passed to the analysis and transformation components of the system, the directed computational graph module **255**, high volume web crawling module **215** and multidimensional time series database **220**. The directed computational graph retrieves one or more streams of data from a plurality of sources, which includes, but is in no way not limited to, a number of physical sensors, web-based questionnaires and surveys, monitoring of electronic infrastructure, crowd sourcing campaigns, and human input device information. Within the directed computational graph, data may be split into two identical streams, wherein one sub-stream may be sent for batch processing and storage while the other sub-stream may be reformatted for transformation pipeline analysis. The data is then transferred to general transformer service **260** for linear data transformation as part of analysis or decomposable transformer service **250** for branching or iterative transformations that are part of analysis. The directed computational graph **255** represents all data as directed graphs where the transformations are nodes and the result messages between transformations edges of the graph. These graphs which contain considerable intermediate transformation data are stored and further analyzed within graph stack module **245**. High volume web crawling module **215** uses multiple server hosted preprogrammed web spiders to find and retrieve data of interest from web-based sources that are not well tagged by conventional web crawling technology. Multiple dimension time series database module **220** receives data from a large plurality of sensors that may be of several different types. The module is designed to accommodate irregular and high volume surges by dynamically allotting network bandwidth and server processing channels to process the incoming data. Data retrieved by the multidimensional time series database **220** and the high volume web crawling module **215** may be further analyzed and transformed into task optimized results by the directed computational graph **255** and associated general transformer service **250** and decomposable transformer service **260** modules.

(33) Results of the transformative analysis process may then be combined with further client directives, additional business rules and practices relevant to the analysis and situational information external to the already available data in the automated planning service module **230** which also runs powerful predictive statistics functions and machine learning algorithms to allow future trends and outcomes to be rapidly forecast based upon the current system derived results and choosing each a plurality of possible business decisions. Using all available data, the automated planning service module **230** may propose business decisions most likely to result is the most favorable business outcome with a usably high level of certainty. Closely related to the automated planning service module in the use of system derived results in conjunction with possible externally supplied additional information in the assistance of end user business decision making, the business outcome simulation module **225** coupled with the end user facing observation and state estimation service **240** allows business decision makers to investigate the probable outcomes of choosing one pending course of action over another based upon analysis of the current available data. For example, the pipelines operations department has reported a very small reduction in crude oil pressure in a section of pipeline in a highly remote section of territory. Many believe the issue is entirely due to a fouled, possibly failing flow sensor, others believe that it is a proximal upstream pump that may have foreign material stuck in it. Correction of both of these possibilities is to

increase the output of the effected pump to hopefully clean out it or the fouled sensor. A failing sensor will have to be replaced at the next maintenance cycle. A few, however, feel that the pressure drop is due to a break in the pipeline, probably small at this point, but even so, crude oil is leaking and the remedy for the fouled sensor or pump option could make the leak much worse and waste much time afterwards. The company does have a contractor about 8 hours away or could rent satellite time to look but both of those are expensive for a probable sensor issue, significantly less than cleaning up an oil spill though and then with significant negative public exposure. These sensor issues have happened before and the business operating system **200** has data from them, which no one really studied due to the great volume of columnar figures, so the alternative courses **225, 240** of action are run. The system, based on all available data predicts that the fouled sensor or pump are unlikely the root cause this time due to other available data and the contractor is dispatched. She finds a small breach in the pipeline. There will be a small cleanup and the pipeline needs to be shutdown for repair but multiple tens of millions of dollars have been saved. This is just one example of a great many of the possible use of the business operating system, those knowledgeable in the art will easily formulate more.

(34) FIG. **3** is a diagram of an exemplary architecture of an automated planning service module and related modules **300** according to an embodiment of the invention. Seen here is a more detailed view of the automated planning service module **230** as depicted in FIG. **2**. The module functions by receiving business decision or business venture candidates as well as relevant currently available related data and any campaign analysis modification commands through a client interface **305**. The module may also be used provide transformed data or run parameters to the action outcome simulation module **225** to seed a simulation prior to run or to transform intermediate result data isolated from one or more actors operating in the action outcome simulation module **225**, during a simulation run. Significant amounts of supporting information such as, but not restricted to current business conditions, infrastructure, ongoing venture status, financial status, market conditions, and world events which may impact the current decision or venture that have been collected by the business operating system as a whole and stored in such data stores as the multidimensional times series database **220**, the analysis capabilities of the directed computational graph module **255** and web-based data retrieval abilities of the high volume web crawler module **215** all of which may be stored in one or more data stores **320, 325** may also be used during simulation of alternative business decision progression, which may entail such variables as, but are not limited to implementation timing, method to end changes, order and timing of constituent part completion or impact of choosing another goal instead of an action currently under analysis.

(35) Contemplated actions may be broken up into a plurality of constituent events that either act towards the fulfillment of the venture under analysis or represent the absence of each event by the discrete event simulation module **311** which then makes each of those events available for information theory based statistical analysis **312**, which allows the current decision events to be analyzed in light of similar events under conditions of varying dis-similarity using machine learned criteria obtained from that previous data; results of this analysis in addition to other factors may be analyzed by an uncertainty estimation module **313** to further tune the level of confidence to be included with the finished analysis. Confidence level would be a weighted calculation of the random variable distribution given to each event analyzed. Prediction of the effects of at least a portion of the events involved with a business venture under analysis within a system as complex as anything from the microenvironment in which the client business operates to more expansive arenas as the regional economy or further, from the perspective of success of the client business is calculated in dynamic systems extraction and inference module **314**, which use, among other tools algorithms based upon Shannon entropy, Hartley entropy and mutual information transfer metrics.

(36) Of great importance in any business decision or new business venture is the amount of business value that is being placed at risk by choosing one decision over another. Often this value is monetary but it can also be competitive placement, operational efficiency or customer

relationship based, for example: the may be the effects of keeping an older, possibly somewhat malfunctioning customer relationship management system one more quarter instead of replacing it for \$14 million dollars and a subscription fee. This is evidenced via profound failures of legacy information technology during times of crisis like the 2022-2023 Southwest airlines debacle stemming from woefully outdated scheduling and data management technologies. The automated planning service module has the ability predict the outcome of such decisions per value that will be placed at risk using programming based upon the Monte Carlo heuristic model **316** which allows a single “state” estimation of value at risk. It is very difficult to anticipate the amount of computing power that will be needed to complete one or more of these business decision analyses which can vary greatly in individual needs and often are run with several alternatives concurrently. This kind of parametric study via organized sampling or manipulation of both initial conditions and of various perturbations to the system (e.g. in the Southwest case various flight conditions, maintenance issues, temperatures, flight crew challenges, and other potential disruptors) enables intentional exploration to determine situations with outsized impact on desired outcomes as measured against specific objective functions (usually valued over finite time horizons). The invention is therefore designed to run on expandable clusters **315**, in a distributed, modular, and extensible approach, such as, but not exclusively, offerings of Amazon's AWS. Similarly, these analysis jobs may run for many hours to completion and many clients may be anticipating long waits for simple “what if” options which will not affect their business operations in the near term while other clients may have come upon a pressing decision situation where they need alternatives as soon as possible. This is accommodated by the presence of a job queue that allows analysis jobs to be implemented at one of multiple priority levels from low to urgent. In case of a change in more hypothetical analysis jobs to more pressing, job priorities can also be changed during run without loss of progress using the priority based job queue **318**.

(37) Structured plan analysis result data may be stored in either a general purpose automated planning engine executing Action Notation Modeling Language (ANML) scripts for modeling which can be used to prioritize both human and machine-oriented tasks to maximize reward functions over finite time horizons **317** or through the graph-based data store **245**, depending on the specifics of the analysis in complexity and time run. The results of evaluations can drive dynamic resampling of potential perturbations of interest in the parametric study or prompt human overseer interaction for interest in expanding searches based on particular observations.

(38) The results of analyses may be sent to one of two client facing presentation modules, the action outcome simulation module **225** or the more visual simulation capable observation and state estimation module **240** depending on the needs and intended usage of the data by the client.

(39) FIG. 4 is a system diagram illustrating connections between core components of the invention for geo-locating and tracking the status of cyber-physical assets, according to a preferred aspect. A business operating system **410** operates an optimization engine **411**, parametric evaluation engine **412**, and uses abstract data representations **413** including Markov State Models (MSM) **414** and abstract representations of finite state machines **415** to read, modify, and generally operate on data. A business operating system **410** such as this is connected to a network **450**, which may be an intranet, the internet, a local area connection, or any one of many other configurations of networks. Also connected to this network **450** is at least one database **420**, which holds information including a crypto-ledger **421**, an implementation of a blockchain data construct, which will be expounded upon in later figures. Connected to a network **450** is at least one cyber-physical asset **430**, **440**, which may hold any number of sensors or data according to a specific implementation, and have geoJSON **431**, **441** data with which to record their geo-physical location. A cyber-physical asset **430**, **440** may be a delivery crate with a possible plurality of sensors and computers embedded or attached to the crate in some way, or may be an object inside a mundane crate such as a piece of research equipment which may communicate with a business operating system **410** during transit, or may be a stationary object such as research equipment, computer systems, and more, which are

capable of sending status updates at least consisting of geoJSON **431**, **441** information regarding their geophysical location over a network **450**. A business operating system may use a Markov State Model (MSM) **414** as a tool for data representation of the states of cyber-physical assets which send status updates in this way, and may or may not reduce a MSM to a finite state machine representation **415** with or without stochastic elements, according to a preferred aspect. These data representations **413** are useful for visualizing and analyzing current, previous, and possible future states of assets **430**, **440** connected to an operating system **410** over a network **450**.

(40) FIG. 5 is a method diagram illustrating key steps in the communication between cyber-physical assets **430**, **440** and remote servers running a business operating system **410**, according to a preferred aspect. Any relevant sensors or sensing equipment and software must be installed on the asset **510** first, before relevant data can be sent to a business operating system **410**. Such sensors may include a variety of implementations, including temperature sensors, GPS tracking software, accelerometers, or any other sensors and accompanying hardware and software as needed or desired by the user upon implementation of this system. The cyber-physical asset **430**, **440** will maintain, as part of their software involvement in the system, a private key, and the requisite software for communicating with the databases. Notably, in one preferred embodiment, data may be periodically archived to a crypto-ledger **421** implementation **520** using blockchain technology to create an immutable record of what state, interaction, or potentially forecasts of interactions or state were known. This can be valuable to establish immutable records of what might be informing either machine or human actions at various points in time. This can provide risk management benefits as a technical inhibitor to “answer changes” where such auditability is desired but administrative rights or technical acumen might make a database at risk to manipulation in select scenarios. Blockchain technology is essentially a method for secure message sending between network connected devices, often used for the purposes of transaction ledgers and smart contracts, using asymmetric encryption. The cyber-physical asset will be in communication with a business operating system **410** either continuously or at set intervals **530**, depending on individual implementations, according to a preferred aspect. During these communications, the asset will, using the asymmetric encryption in blockchain crypto-ledgers, send status updates based on any sensors installed on the asset **530**. A business operating system that receives these updates will then verify them with previous status updates in databases **540** to ensure that the updates received are legitimate, and not forged or from a dubious source. If the public key, or signature, or contents of the encrypted message are not able to be verified properly, the ledger held in at least one database is not updated **560**. If they are properly verified and indicate they are from the real asset and indicate a legitimate status update, any databases which hold a copy of the crypto-ledger **421** are updated with the new status of the asset **550**. It will be apparent to one skilled in the art that additional uses for an update verification process may be that partial updates (for example, with certain pieces of data not sent to the server in the status update) may be used, and with this partial observability, missing data between status updates may be inferred using machine learning techniques. It is possible to implement a rules engine for this purpose, to determine what rules to apply for inference of missing data, depending on the implementation of the system.

(41) FIG. 6 is a method diagram illustrating key steps in a business operating system **410** interacting with data received from cyber-physical assets **430**, **440** in databases **420** to verify updates in a cryptographic ledger **421**, according to a preferred aspect. Any asset must generate a public and private key **610** in accordance with the specifications of asymmetric encryption, which are known technologies in the art. An asset must prepare an update **620**, which may mean formatting data received from any installed sensors, performing any relevant calculations or modifications to raw data, and preparing any network devices for sending the data across a network **450**. The cyber-physical asset **430**, **440** must sign any update with its private key **630**, which encrypts the update in a way that only the private or public keys can be used to decrypt. The asset, when connected to a network **450**, may send the prepared and encrypted update to any “nodes” or

computer systems running a business operating system **410**, to be verified before being added onto the ledger **421**, **640**. Any nodes running a business operating system **410** will attempt to verify the asset status update **650**, before then verifying with the ledger held in at least one database **420** and any other relevant nodes or computer systems with such a business operating system **410** that the asset update is legitimate, valid, and shall be added to the ledger of status updates from the asset **660**. It is possible to implement this system and method in an ongoing identification and authentication service, for continuous updates, rather than discrete authentication and verification for discrete updates.

(42) FIG. **7** is a method diagram illustrating several steps in the use of smart contracts combined with cyber-physical assets, according to a preferred aspect. Such smart contracts are possible as a result of implementing blockchain technology to not only keep track of and verify entries in crypto-ledgers **421**, but to store and execute distributed programs, for the purposes of self-enforcing contracts, known as smart contracts. In this implementation, a smart contract is implemented with a domain-specific-language (DSL) which may be provided by a vendor of the system or specified by a user of the system **710**. A DSL may be thought of as a custom programming language, and may, depending on the implementation, also be an otherwise unmodified implementation of a programming language, according to a preferred aspect. Conditions for smart contracts in this system may be based on the past, present, or future status of cyber-physical assets monitored by the system **720**. Upon completion of whatever conditions are programmed into a smart contract, the contract program executes, which may perform any number of tasks that may be programmed into a computer, including withdrawal of funds, depositing of funds, messages sent across a network **450**, or other similar results of an executed program **730**, according to a preferred aspect. These parametrically-triggered remuneration contracts may be versatile and diverse in their implementation according to the needs of the consumer. This can be important if decentralized actions are needed. E.g. preloading a smart contract that would authorize payment and resource expenditure for an item of interest whose state or estimated state is outside of desired maintenance or preservation parameters. One representative example is a painting on loan that is being transferred or is on loan to from a museum—if moisture or temperature thresholds are violated it may be desirable to need no further information or approvals to save the work or pre-emptively dispatch resources to combat actual or would-be damage.

(43) FIG. **8** is a method diagram illustrating key steps in the function of a parametric evaluation engine **412**, according to a preferred aspect. A parametric evaluation engine **412** may query at least one database **420** for a ledger **421** containing previous or current status updates of at least one cyber-physical asset **430**, **440**, **810**. This query may be performed across a network **450** from a business operating system **410** run on a computer system and may take the form of any database query format, including NOSQL™ databases such as MONGODB™, or SQL™ databases including MICROSOFT SQL SERVER™ and MYSQL™ databases, depending on the desired database implementation in the system, according to a preferred aspect. Asset status histories may be returned to a parametric evaluation engine **412**, which may be listed to a user of the engine, in a basic user interface which allows the listing and searching of such asset status update histories **820**. Asset statuses may be viewed over time as a history rather than listed separately, if desired, for the purpose of noting and examining trends in an asset's status **830**, according to an aspect.

(44) FIG. **13** is a method diagram illustrating an exemplary process for using status update messages to predict the status of network resources and continuously refine predictive logic, according to a preferred aspect. According to this method, status updates may be received **1310** and monitored **1320** to identify status patterns and trends for network resources such as (for example, including but not limited to) servers, user computing devices such as laptops or smartphones, routers, hardware firewall or VPN devices, network switches, network sensors, Internet of things (IoT) devices, automation controllers, or any other network-connected devices. Status updates may vary in content and may comprise any information that may be useful to report, such as a simple

connectivity heartbeat for an IoT device to confirm online status, or a complex report of detailed firewall configuration from a firewall device. Many network devices are capable of reporting logs that are generated in a standardized format as part of normal operation using reporting tools such as (for example, including but not limited to) rsyslog or similar system utilities that are able to export automatically-generated logs over a network connection in a consistent format for storage and use. A preferred embodiment of the invention is to leverage mTLS for transport, but other techniques including site-to-site VPNs or basic https calls may also be used.

(45) Tracked status trends may then be analyzed and extrapolated **1330** to predict future states of and changes to network resources. Over time, new status updates may or may not be received **1340** based on a variety of conditions such as (for example) device connectivity or network issues. If an update is not received when expected **1350**, a predicted status may be assumed until such time as an update is received with the actual status for a resource. This enables operations to continue without interruption, by proceeding with the predicted likely status trend until the actual status can be confirmed and operation adjusted to match. When a status update is received, it may be compared against the predicted status **1360** in order to analyze the difference between the predicted values and the real ones. This may then be used to refine future predictions **1370**, continuously improving operation to maintain relevant predicted values with a high degree of confidence so that gaps in status reporting do not interfere with operations.

(46) FIG. **14** is a method diagram illustrating an exemplary method for scheduling network tasks based on predicted availability and conditions, according to a preferred aspect. According to this method, the status of network resources may be monitored over time **1410** to track status patterns and trends, as described above (with reference to FIG. **13**). When a network task is received **1420**, such as (for example, including but not limited to) a configuration change, software update, or maintenance task, the appropriate network resource's availability (based on the task received, for example the resource intended to be modified or a resource appropriate for handling the particular task) may be checked **1430** to determine whether the resource is ready to process the task immediately. If the resource is available, the task may be assigned for immediate handling **1440**. If, however, the resource is currently unavailable (such as being offline, status unknown, or already under sufficient load that the task cannot be accommodated at the current time), the resource's predicted status (as determined according to the method described previously, with reference to FIG. **13**) may be checked **1450** to determine future availability for handling the task. The task may then be scheduled for assignment **1460** to be handled by the network resource at an appropriate future time based on the predicted status and availability.

(47) Hardware Architecture

(48) Generally, the techniques disclosed herein may be implemented on hardware or a combination of software and hardware. For example, they may be implemented in an operating system kernel, in a separate user process, in a library package bound into network applications, on a specially constructed machine, on an application-specific integrated circuit (“ASIC”), or on a network interface card.

(49) Software/hardware hybrid implementations of at least some of the aspects disclosed herein may be implemented on a programmable network-resident machine (which should be understood to include intermittently connected network-aware machines) selectively activated or reconfigured by a computer program stored in memory. Such network devices may have multiple network interfaces that may be configured or designed to utilize different types of network communication protocols. A general architecture for some of these machines may be described herein in order to illustrate one or more exemplary means by which a given unit of functionality may be implemented. According to specific aspects, at least some of the features or functionalities of the various aspects disclosed herein may be implemented on one or more general-purpose computers associated with one or more networks, such as for example an end-user computer system, a client computer, a network server or other server system, a mobile computing device (e.g., tablet

computing device, mobile phone, smartphone, laptop, or other appropriate computing device), a consumer electronic device, a music player, or any other suitable electronic device, router, switch, or other suitable device, or any combination thereof. In at least some aspects, at least some of the features or functionalities of the various aspects disclosed herein may be implemented in one or more virtualized computing environments (e.g., network computing clouds, virtual machines hosted on one or more physical computing machines, or other appropriate virtual environments).

(50) Referring now to FIG. 9, there is shown a block diagram depicting an exemplary computing device **10** suitable for implementing at least a portion of the features or functionalities disclosed herein. Computing device **10** may be, for example, any one of the computing machines listed in the previous paragraph, or indeed any other electronic device capable of executing software- or hardware-based instructions according to one or more programs stored in memory. Computing device **10** may be configured to communicate with a plurality of other computing devices, such as clients or servers, over communications networks such as a wide area network a metropolitan area network, a local area network, a wireless network, the Internet, or any other network, using known protocols for such communication, whether wireless or wired.

(51) In one embodiment, computing device **10** includes one or more central processing units (CPU) **12**, one or more interfaces **15**, and one or more busses **14** (such as a peripheral component interconnect (PCI) bus). When acting under the control of appropriate software or firmware, CPU **12** may be responsible for implementing specific functions associated with the functions of a specifically configured computing device or machine. For example, in at least one embodiment, a computing device **10** may be configured or designed to function as a server system utilizing CPU **12**, local memory **11** and/or remote memory **16**, and interface(s) **15**. In at least one embodiment, CPU **12** may be caused to perform one or more of the different types of functions and/or operations under the control of software modules or components, which for example, may include an operating system and any appropriate applications software, drivers, and the like.

(52) CPU **12** may include one or more processors **13** such as, for example, a processor from one of the Intel, ARM, Qualcomm, and AMD families of microprocessors. In some embodiments, processors **13** may include specially designed hardware such as application-specific integrated circuits (ASICs), electrically erasable programmable read-only memories (EEPROMs), field-programmable gate arrays (FPGAs), and so forth, for controlling operations of computing device **10**. In a specific embodiment, a local memory **11** (such as non-volatile random access memory (RAM) and/or read-only memory (ROM), including for example one or more levels of cached memory) may also form part of CPU **12**. However, there are many different ways in which memory may be coupled to system **10**. Memory **11** may be used for a variety of purposes such as, for example, caching and/or storing data, programming instructions, and the like. It should be further appreciated that CPU **12** may be one of a variety of system-on-a-chip (SOC) type hardware that may include additional hardware such as memory or graphics processing chips, such as a QUALCOMM SNAPDRAGON™ or SAMSUNG EXYNOS™ CPU as are becoming increasingly common in the art, such as for use in mobile devices or integrated devices.

(53) As used herein, the term “processor” is not limited merely to those integrated circuits referred to in the art as a processor, a mobile processor, or a microprocessor, but broadly refers to a microcontroller, a microcomputer, a programmable logic controller, an application-specific integrated circuit, and any other programmable circuit. In one embodiment, interfaces **15** are provided as network interface cards (NICs).

(54) Generally, NICs control the sending and receiving of data packets over a computer network; other types of interfaces **15** may for example support other peripherals used with computing device **10**. Among the interfaces that may be provided are Ethernet interfaces, frame relay interfaces, cable interfaces, DSL interfaces, token ring interfaces, graphics interfaces, and the like. In addition, various types of interfaces may be provided such as, for example, universal serial bus (USB), Serial, Ethernet, FIREWIRE™, THUNDERBOLT™, PCI, parallel, radio frequency (RF),

BLUETOOTH™, near-field communications (e.g., using near-field magnetics), 802.11 (WiFi), frame relay, TCP/IP, ISDN, fast Ethernet interfaces, Gigabit Ethernet interfaces, Serial ATA (SATA) or external SATA (ESATA) interfaces, high-definition multimedia interface (HDMI), digital visual interface (DVI), analog or digital audio interfaces, asynchronous transfer mode (ATM) interfaces, high-speed serial interface (HSSI) interfaces, Point of Sale (POS) interfaces, fiber data distributed interfaces (FDDIs), and the like. Generally, such interfaces **15** may include physical ports appropriate for communication with appropriate media. In some cases, they may also include an independent processor (such as a dedicated audio or video processor, as is common in the art for high-fidelity A/V hardware interfaces) and, in some instances, volatile and/or non-volatile memory (e.g., RAM).

(55) Although the system shown in FIG. **9** illustrates one specific architecture for a computing device **10** for implementing one or more of the inventions described herein, it is by no means the only device architecture on which at least a portion of the features and techniques described herein may be implemented. For example, architectures having one or any number of processors **13** may be used, and such processors **13** may be present in a single device or distributed among any number of devices. In one embodiment, a single processor **13** handles communications as well as routing computations, while in other embodiments a separate dedicated communications processor may be provided. In various embodiments, different types of features or functionalities may be implemented in a system according to the invention that includes a client device (such as a tablet device or smartphone running client software) and server systems (such as a server system described in more detail below).

(56) Regardless of network device configuration, the system of the present invention may employ one or more memories or memory modules (such as, for example, remote memory block **16** and local memory **11**) configured to store data, program instructions for the general-purpose network operations, or other information relating to the functionality of the embodiments described herein (or any combinations of the above). Program instructions may control execution of or comprise an operating system and/or one or more applications, for example. Memory **16** or memories **11**, **16** may also be configured to store data structures, configuration data, encryption data, historical system operations information, or any other specific or generic non-program information described herein.

(57) Because such information and program instructions may be employed to implement one or more systems or methods described herein, at least some network device embodiments may include nontransitory machine-readable storage media, which, for example, may be configured or designed to store program instructions, state information, and the like for performing various operations described herein. Examples of such nontransitory machine-readable storage media include, but are not limited to, magnetic media such as hard disks, floppy disks, and magnetic tape; optical media such as CD-ROM disks; magneto-optical media such as optical disks, and hardware devices that are specially configured to store and perform program instructions, such as read-only memory devices (ROM), flash memory (as is common in mobile devices and integrated systems), solid state drives (SSD) and “hybrid SSD” storage drives that may combine physical components of solid state and hard disk drives in a single hardware device (as are becoming increasingly common in the art with regard to personal computers), memristor memory, random access memory (RAM), and the like. It should be appreciated that such storage means may be integral and non-removable (such as RAM hardware modules that may be soldered onto a motherboard or otherwise integrated into an electronic device), or they may be removable such as swappable flash memory modules (such as “thumb drives” or other removable media designed for rapidly exchanging physical storage devices), “hot-swappable” hard disk drives or solid state drives, removable optical storage discs, or other such removable media, and that such integral and removable storage media may be utilized interchangeably. Examples of program instructions include both object code, such as may be produced by a compiler, machine code, such as may be produced by an assembler or a linker, byte

code, such as may be generated by for example a JAVA™ compiler and may be executed using a Java virtual machine or equivalent, or files containing higher level code that may be executed by the computer using an interpreter (for example, scripts written in Python, Perl, Ruby, Groovy, or any other scripting language).

(58) In some embodiments, systems according to the present invention may be implemented on a standalone computing system. Referring now to FIG. 10, there is shown a block diagram depicting a typical exemplary architecture of one or more embodiments or components thereof on a standalone computing system. Computing device 20 includes processors 21 that may run software that carry out one or more functions or applications of embodiments of the invention, such as for example a client application 24. Processors 21 may carry out computing instructions under control of an operating system 22 such as, for example, a version of MICROSOFT WINDOWS™ operating system, APPLE OSX™ or iOS™ operating systems, some variety of the Linux operating system, ANDROID™ operating system, or the like. In many cases, one or more shared services 23 may be operable in system 20, and may be useful for providing common services to client applications 24. Services 23 may for example be WINDOWS™ services, user-space common services in a Linux environment, or any other type of common service architecture used with operating system 21. Input devices 28 may be of any type suitable for receiving user input, including for example a keyboard, touchscreen, microphone (for example, for voice input), mouse, touchpad, trackball, or any combination thereof. Output devices 27 may be of any type suitable for providing output to one or more users, whether remote or local to system 20, and may include for example one or more screens for visual output, speakers, printers, or any combination thereof. Memory 25 may be random-access memory having any structure and architecture known in the art, for use by processors 21, for example to run software. Storage devices 26 may be any magnetic, optical, mechanical, memristor, or electrical storage device for storage of data in digital form (such as those described above, referring to FIG. 9). Examples of storage devices 26 include flash memory, magnetic hard drive, CD-ROM, and/or the like.

(59) In some embodiments, systems of the present invention may be implemented on a distributed computing network, such as one having any number of clients and/or servers. Referring now to FIG. 11, there is shown a block diagram depicting an exemplary architecture 30 for implementing at least a portion of a system according to an embodiment of the invention on a distributed computing network. According to the embodiment, any number of clients 33 may be provided. Each client 33 may run software for implementing client-side portions of the present invention; clients may comprise a system 20 such as that illustrated in FIG. 10. In addition, any number of servers 32 may be provided for handling requests received from one or more clients 33. Clients 33 and servers 32 may communicate with one another via one or more electronic networks 31, which may be in various embodiments any of the Internet, a wide area network, a mobile telephony network (such as CDMA or GSM cellular networks), a wireless network (such as WiFi, WiMAX, LTE, and so forth), or a local area network (or indeed any network topology known in the art; the invention does not prefer any one network topology over any other). Networks 31 may be implemented using any known network protocols, including for example wired and/or wireless protocols.

(60) In addition, in some embodiments, servers 32 may call external services 37 when needed to obtain additional information, or to refer to additional data concerning a particular call. Communications with external services 37 may take place, for example, via one or more networks 31. In various embodiments, external services 37 may comprise web-enabled services or functionality related to or installed on the hardware device itself. For example, in an embodiment where client applications 24 are implemented on a smartphone or other electronic device, client applications 24 may obtain information stored in a server system 32 in the cloud or on an external service 37 deployed on one or more of a particular enterprise's or user's premises.

(61) In some embodiments of the invention, clients 33 or servers 32 (or both) may make use of one

or more specialized services or appliances that may be deployed locally or remotely across one or more networks **31**. For example, one or more databases **34** may be used or referred to by one or more embodiments of the invention. It should be understood by one having ordinary skill in the art that databases **34** may be arranged in a wide variety of architectures and using a wide variety of data access and manipulation means. For example, in various embodiments one or more databases **34** may comprise a relational database system using a structured query language (SQL), while others may comprise an alternative data storage technology such as those referred to in the art as “NoSQL” (for example, HADOOP CASSANDRA™, GOOGLE BIGTABLE™, and so forth). In some embodiments, variant database architectures such as column-oriented databases, in-memory databases, clustered databases, distributed databases, or even flat file data repositories may be used according to the invention. It will be appreciated by one having ordinary skill in the art that any combination of known or future database technologies may be used as appropriate, unless a specific database technology or a specific arrangement of components is specified for a particular embodiment herein. Moreover, it should be appreciated that the term “database” as used herein may refer to a physical database machine, a cluster of machines acting as a single database system, or a logical database within an overall database management system. Unless a specific meaning is specified for a given use of the term “database”, it should be construed to mean any of these senses of the word, all of which are understood as a plain meaning of the term “database” by those having ordinary skill in the art.

(62) Similarly, most embodiments of the invention may make use of one or more security systems **36** and configuration systems **35**. Security and configuration management are common information technology (IT) and web functions, and some amount of each are generally associated with any IT or web systems. It should be understood by one having ordinary skill in the art that any configuration or security subsystems known in the art now or in the future may be used in conjunction with embodiments of the invention without limitation, unless a specific security **36** or configuration system **35** or approach is specifically required by the description of any specific embodiment.

(63) FIG. **12** shows an exemplary overview of a computer system **40** as may be used in any of the various locations throughout the system. It is exemplary of any computer that may execute code to process data. Various modifications and changes may be made to computer system **40** without departing from the broader scope of the system and method disclosed herein. Central processor unit (CPU) **41** is connected to bus **42**, to which bus is also connected memory **43**, nonvolatile memory **44**, display **47**, input/output (I/O) unit **48**, and network interface card (NIC) **53**. I/O unit **48** may, typically, be connected to keyboard **49**, pointing device **50**, hard disk **52**, and real-time clock **51**. NIC **53** connects to network **54**, which may be the Internet or a local network, which local network may or may not have connections to the Internet. Also shown as part of system **40** is power supply unit **45** connected, in this example, to a main alternating current (AC) supply **46**. Not shown are batteries that could be present, and many other devices and modifications that are well known but are not applicable to the specific novel functions of the current system and method disclosed herein. It should be appreciated that some or all components illustrated may be combined, such as in various integrated applications, for example Qualcomm or Samsung system-on-a-chip (SOC) devices, or whenever it may be appropriate to combine multiple capabilities or functions into a single hardware device (for instance, in mobile devices such as smartphones, video game consoles, in-vehicle computer systems such as navigation or multimedia systems in automobiles, or other integrated hardware devices).

(64) In various embodiments, functionality for implementing systems or methods of the present invention may be distributed among any number of client and/or server components. For example, various software modules may be implemented for performing various functions in connection with the present invention, and such modules may be variously implemented to run on server and/or client components.

(65) The skilled person will be aware of a range of possible modifications of the various embodiments described above. Accordingly, the present invention is defined by the claims and their equivalents.

Claims

1. A system for predictive cyber-physical resource management, comprising: a first computing device comprising a first processor, a first memory, a wired or wireless communication device, and a first plurality of programming instructions stored in the first memory and operating on the first processor, wherein the first plurality of programmable instructions, when operating on the first processor, cause the computing device to: receive status update messages from one or more physical assets; for each physical asset for which a status message is received, update a current state record for the physical asset based on the content of the status message; append the current state record to a time-series record of the asset's state over time; where a status update message for a particular physical asset is not received, apply a machine learning algorithm to the state model to determine a predicted state of the particular physical asset; and append the predicted state to a time-series record of the particular asset's state over time.
2. The system of claim 1, further comprising a plurality of second computing devices each attached to one of a plurality of physical assets, each second computing device comprising a second processor, a second memory, a device identifier, a geolocation device, a wireless communication device, and a second plurality of programming instructions stored in the second memory and operating on the first processor, wherein the second plurality of programmable instructions, when operating on the second processor, cause the second computing device to: periodically determine a state of the physical asset to which it is attached using the geolocation device of the first computing device, the state comprising one or more attributes of the physical asset; periodically generate a status update message, the status update message comprising the device identifier of the first computing device and the determined state of the physical asset to which the first computing device is attached; send the status update message via the wireless communication device to the second computing device.
3. The system of claim 1, further comprising a non-volatile data storage device, wherein databases stored on the non-volatile data storage device comprise a multi-dimensional time series database and a location database.
4. The system of claim 3, wherein the location database is a decentralized, peer-to-peer blockchain database.
5. The system of claim 1, further comprising a parametric evaluation engine comprising a third plurality of programming instructions stored in the second memory and operating on the second processor, wherein the third plurality of programmable instructions, when operating on the second processor, cause the second computing device to: generate a smart contract for the physical asset; receive a parameter for execution of the smart contract, the parameter specifying a condition for execution of the contract; monitor the status update messages for the parameter; and execute the smart contract when the parameter is detected.
6. The system of claim 1, wherein the first computing device is further configured to: receive a task for handling by a physical asset; select a physical asset to handle the received task; determine an availability of the physical asset based on the time-series record of the asset's state over time; and schedule the task for handling based on the determined availability of the physical asset.
7. The system of claim 6, wherein the task is scheduled for immediate handling.
8. The system of claim 6, wherein the task is scheduled for future handling based on a predicted state of the selected physical asset.
9. A method for predictive cyber-physical resource management, comprising the steps of: receiving, at a first computing device, status update messages from one or more physical assets; for

each physical asset for which a status message is received, updating a current state record for the physical asset based on the content of the status message; appending the current state record to a time-series record of the asset's state over time; where a status update message for a particular physical asset is not received, applying a machine learning algorithm to the state model to determine a predicted state of the particular physical asset; and appending the predicted state to a time-series record of the particular asset's state over time.

10. The method of claim 9, further comprising the steps of: periodically determining, using a second computing device, a state of the physical asset to which it is attached using the geolocation device of the first computing device, the state comprising one or more attributes of the physical asset; periodically generating a status update message, the status update message comprising the device identifier of the first computing device and the determined state of the physical asset to which the first computing device is attached; sending the status update message via the wireless communication device to the second computing device.

11. The method of claim 9, further comprising a non-volatile data storage device, wherein databases stored on the non-volatile data storage device comprise a multi-dimensional time series database and a location database.

12. The method of claim 11, wherein the location database is a decentralized, peer-to-peer blockchain database.

13. The method of claim 9, further comprising the steps of: generating, at a parametric evaluation engine comprising a third computing device, a smart contract for the physical asset; receiving a parameter for execution of the smart contract, the parameter specifying a condition for execution of the contract; monitoring the status update messages for the parameter; and executing the smart contract when the parameter is detected.

14. The method of claim 9, further comprising the steps of: receiving a task for handling by a physical asset; selecting a physical asset to handle the received task; determining an availability of the physical asset based on the time-series record of the asset's state over time; and scheduling the task for handling based on the determined availability of the physical asset.

15. The method of claim 14, wherein the task is scheduled for immediate handling.

16. The method of claim 14, wherein the task is scheduled for future handling based on a predicted state of the selected physical asset.
